

Enhancing Functional Brain Network Construction for Neurological Disorder Diagnosis through Modularity-Driven Regularization

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Abstract—The burgeoning field of Functional Brain Network (FBN) analysis has emerged as a promising avenue for diagnosing neurological disorders. As innovative biomarkers, FBNs offer a novel perspective on probing the intricacies of brain connectivity. However, the fidelity of FBN construction depends on the quality of data acquisition, which is often a formidable challenge due to inherent limitations. Inspired by the *no free lunch* theorem, a straightforward ambition is to integrate the exogenous knowledge to enhance the fidelity of FBNs. Motivated by this ambition, this paper provides a Modularity-Driven Regularization (MDR) for the FBN construction. Specifically, we design and incorporate a dyadic L_{21} regulariser into the FBN model to formulate the modularity prior. Furthermore, we have developed a generalized alternating multiplier method algorithm for the estimation of the FBN. To demonstrate the effectiveness of our MDR, we perform rigorous experiments on a Mild Cognitive Impairment (MCI) identification task using the NITRC dataset. The validity of our model has been confirmed by the experimental outcomes. Additionally, it enhances biomarker precision for diagnosing neurological disorders like MCI, supporting potential early intervention strategies.

Index terms—Functional Brain Networks (FBN), Modularity, Mild Cognitive Impairment, Generalized Alternating Multiplier Method Algorithm

I. INTRODUCTION

The human brain is one of the most complex systems in the known universe, comprising approximately 86 billion nerve cells and trillions of synaptic connections [1]. Given the intricate nature of its structure and function, investigating brain operation and information transfer mechanisms has become a pivotal area

of scientific inquiry. A deeper understanding of these mechanisms is essential for unraveling the mysteries of human cognition, perception, behavior, and the complexities underlying neurological disorders.

Functional Brain Networks (FBNs) have become an indispensable technical means in the diagnosis and treatment of neurology. As groundbreaking biomarkers, FBNs offer new opportunities for exploring brain connectivity and identifying patterns indicative of specific diseases. This has profound implications for the management of neuropsychiatric conditions such as depression [2, 3], Parkinson's disease [4, 5], autism [6-9], and schizophrenia [10]. Naturally, constructing FBNs is a key step in understanding the functional connectivity of the brain.

To better construct brain networks, a variety of brain network modeling methods have been proposed for the construction of FBNs. These include Pearson correlation-based models [11], partial correlation models [12], Bayesian network models, structural equation models [13], dynamic causal models [14], among others. Each method strives to balance statistical fidelity (i.e., accuracy in representing true functional connectivity), computational robustness, and physiological relevance in its own way. The success of these methods hinges on the quality of the brain network they aim to represent, which in turn is heavily influenced by the initial acquisition and preprocessing of neuroimaging data. The fidelity of brain network construction is intricately linked to the purity and consistency of functional magnetic resonance imaging (fMRI) data [15].

Regrettably, observed data are often marred by artifacts or noise, and improper processing steps, such as temporal correction and spatial normalization can introduce further distortions. These issues can severely compromise the final brain network model, leading to analyses that are less than reliable [16]. Existing methods attempt to reduce noise by incorporating a priori information, such as sparsity [17], small-world property [18], and scale-free property [19]. Nevertheless,

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the intricacies of brain networks pose a formidable challenge in effectively harnessing their topological configurations. Consequently, the design of regularization schemes that accurately capture the attributes of these complex graphs remains a significant endeavor in the field.

Therefore, to elevate the quality of brain network construction, we have introduced the concept of modularity into the construction of brain networks. Modularity streamlines network architecture and elucidates functional organization by segmenting the network into multiple modules characterized by dense internal connections and sparse external linkages [20]. This modular framework is particularly crucial for discerning the intrinsic patterns and dynamics of brain activity, as many neurological disorders are marked by significant shifts in the functional connectivity between brain regions.

Further, to formulate the modular prior, we introduce a modularity-based brain network modelling framework that enhances the fidelity of FBN by integrating exogenous knowledge, especially modular information. In this context, exogenous knowledge refers to a priori knowledge or information from external sources, such as previous biological studies or pre-existing brain region delineations. We construct a modular prior by designing and incorporating a binary $L_{2,1}$ regulariser into the FBN construction model. This innovative approach offers a more precise representation of functional connectivity among brain regions, strengthens the model's sparsity and robustness, and effectively handles noisy data while integrating biological a priori knowledge. Furthermore, we present a Generalized Alternating Direction Method of Multipliers (GADMM) algorithm for the estimation of FBNs, marking a significant advancement in the field of brain network analysis.

To validate the effectiveness of our proposed framework, we conducted rigorous experiments on the Mild Cognitive Impairment (MCI) recognition task using a dataset derived from the Neuroimaging Informatics Tools and Resources Clearinghouse (NITRC). We employ Support Vector Machines (SVM) for brain network classification, aiming to improve classification accuracy and generalization for neurological disease prediction. The experimental results validate our model's effectiveness in accurately identifying MCI, highlighting its potential for clinical applications. The contributions of this paper are demonstrated in the following key areas:

1. Convex objective function: MDR corresponds to a benign convex objective function, which can be efficiently optimized using GADMM;
2. Integration of modular information: binary $L_{2,1}$ regulariser is introduced to incorporate modular information into the FBN construction model to enhance the fidelity of the brain network;
3. Improvement of robustness and sparsity: a modelling framework is developed to better handle noisy data, enhance sparsity and improve the robustness of brain network models.

II. RELATED WORKS

The Pearson correlation-based brain network model is widely used by researchers due to its simplicity and intuitive

approach [13]. This method assesses functional connectivity between brain regions by computing the linear correlation of their blood oxygenation level dependent (BOLD) signals.

First, we assume that the brain is divided into n brain regions (ROI) through some atlas, and $\mathbf{X} = [x_1, x_2, \dots, x_n] \in R^{m \times n}$ is the data matrix of fMRI corresponding to all the brain regions we obtained, where $x_i \in R^m (i = 1, \dots, n)$ is the sequence of BOLD signals in the i brain region, m is the number of time points, and n is the number of subjects.

A. Pearson's Correlation

Then in the Pearson correlation-based brain network model, the weight edge value w_{ij} between the i brain region and the j brain region is defined as follows:

$$w_{ij} = \frac{(x_i - \bar{x}_i)^T (x_j - \bar{x}_j)}{\sqrt{(x_i - \bar{x}_i)^T (x_i - \bar{x}_i)} \sqrt{(x_j - \bar{x}_j)^T (x_j - \bar{x}_j)}} \quad (1)$$

where $\bar{x}_i \in R^m (i = 1, 2, \dots, n)$ denotes the mean value corresponding to all x_i . Normalising x_i , i.e. $(x_i - \bar{x}_i)$, and simplifying the formula to $w_{ij} = x_i x_j$, the optimal solution corresponds to the following model:

$$\min_{w_{ij}} \sum_{i,j=1}^n \|x_i - w_{ij} x_j\|^2, \quad (2)$$

Equation (2) is next reduced to the following matrix form:

$$\min_W \|\mathbf{W} - \mathbf{X}^T \mathbf{X}\|_F^2, \quad (3)$$

where $\mathbf{W} = (w_{ij}) \in R^{n \times n}$ is the estimated brain functional network side weight matrix and $\|\cdot\|_F$ is the F-paradigm of the matrix.

B. Partial Correlation Model

Partial correlation assesses whether two nodes are directly linked after eliminating the influence of other nodes. By eliminating the influence of other nodes, the correlation between these two nodes can be estimated more accurately. A zero value of partial correlation means that there is no direct connectivity between two nodes.

Partial correlation between nodes i and j is defined as the correlation between X_i and X_j conditioning on all the other nodes, i.e.:

$$\rho_{ij} = \text{corr}(X_i, X_j | X_{-(i,j)}), X_{-(i,j)} = (X_k; 1 \leq k \neq i, j \leq M) \\ i, j = 1, \dots, M, i \neq j.$$

Partial correlation can be derived not only through linear regression but also by calculating the inverse of the covariance matrix, known as the precision matrix.

Let Σ be the X -based $M \times M$ covariance matrix and let $\Omega = \Sigma^{-1} = \{\omega_{ij}\}_{M \times M}$ be the precision matrix. The partial correlation between nodes i and j can then be expressed as

$$\rho_{ij} = \frac{-\omega_{ij}}{\sqrt{\omega_{ii}\omega_{jj}}}. \quad (4)$$

Under the assumption that the data follows a Gaussian distribution, ρ_{ij} equals 0 implies conditional independence between nodes i and j given all other nodes. As such, partial correlation offers a robust method for identifying direct connectivity and accurately estimating the network by removing the influence of confounding variables.

C. Partial Correlation Based on Sparse Expression

Due to the inevitable noise interference during fMRI data acquisition, the Pearson correlation-based brain network model tends to produce many spurious connections, and therefore a thresholding process is required to filter this connection. Although this processing method is simple and direct, the selection of the threshold value often relies on the researcher's personal experience and subjective judgement, and lacks a certain degree of objectivity and reproducibility. In addition, the thresholding process may also lead to some important connections being mistakenly deleted, thus affecting the accuracy of brain network analysis.

To address the limitations of the Pearson correlation-based method, researchers proposed a biased correlation-based brain network construction method. This method solves the confusion problem due to different nodes interacting with each other by approximating the form of the inverse of the full correlation (i.e., Pearson correlation) [14]. However, the partial order correlation model often encounters the singular solution problem during the solution process, leading to some limitations of the model in practical applications. To address this issue, researchers have obtained stable solutions by introducing regularization terms to constrain the solution space. For example, we can introduce the L_1 -paradigm as a regularization term into the framework of brain networks. The most common concept of Sparse Representation (SR) is also obtained, which is mathematically modelled as:

$$\min_w \sum_{i=1}^n \left(\|x_i - \sum_{j \neq i} w_{ij} x_j\|_2^2 + \lambda_1 \sum_{j \neq i} |w_{ij}| \right),$$

$$s. t. w_{ii} = 0, \forall i = 1, 2, \dots, n, \quad (5)$$

Reducing equation (5) to:

$$\min_w \|\mathbf{X} - \mathbf{X}\mathbf{W}\|_F^2 + \lambda_1 \|\mathbf{W}\|_1,$$

$$s. t. w_{ii} = 0, \forall i = 1, 2, \dots, n, \quad (6)$$

The SR-based partial correlation brain network modeling method finds partial correlations between brain regions by minimizing an objective function. This function includes a data fitting term $\|\mathbf{X} - \mathbf{X}\mathbf{W}\|_F^2$, which approximates the real relationship between brain regions, and an L_1 -norm regularization term to control solution sparsity and stability. By adjusting the regularization parameter λ_1 , the sparsity of the functional brain network can be controlled. And the use of $w_{ii} = 0$ is to remove x_i from $\mathbf{X}$ to avoid equation (6) from appearing as a plain.

III. MODULARITY-DRIVEN REGULARIZATION

A. Modularity-Driven Regularization for FBN Construction

The traditional biased correlation-based brain network framework model is first given:

$$\mathbf{X} = \mathbf{X}\mathbf{W}, \quad (7)$$

where $\mathbf{X} = [x_1, x_2, \dots, x_n] \in R^{d \times n}$ is the time series of ROIs for each brain region and n represents the number of time points. To model the acquisition of a block diagonal representation matrix as an equivalent problem of discovering the ranks of associations in that matrix. Next the corresponding $L_{2,1}$ paradigm is embedded into the data fitting term to

simultaneously shrink the ranks of the representation matrix and the objective function of the model is formalised as.

$$\min_{\mathbf{W}} \frac{1}{2} \|\mathbf{X} - \mathbf{X}\mathbf{W}\|_F^2 + \lambda R(\mathbf{W}). \quad (8)$$

$R(\mathbf{W}) = \sum_{(i,j) \in E} \eta_{ij} \|w_i - w_j\|_2 + \sum_{(i,j) \in E} \eta_{ij} \|w_i - w_j\|_2$ is convex with respect to $\mathbf{W}$, E is the set of edges formed by the set of edges formed by the fMRI data point pairs based on the similarity of the BOLD signals, $\mathbf{W}_i(\mathbf{W}_i)$ is the i (i column) of the coefficient matrix, and η_{ij} is the non-negative weight coefficient, which indicates the similarity between two data points. Since the columns and their corresponding rows represent new forms of the original data, the row and column fusion terms share identical weights, denoted as η_{ij} . Referring to [22], in (8), we use sparse Gaussian mixture weights $\eta_{ij} = \iota_{(i,j)}^k \exp(-\phi \|\mathbf{X}_i - \mathbf{X}_j\|_2^2)$. where $\iota_{(i,j)}^k$ is the indicator function, where if $\mathbf{X}_i$ is one of the k -nearest neighbours of $\mathbf{X}_j$ the corresponding indicator value is 1, and 0 otherwise. $\exp(-\phi \|\mathbf{X}_i - \mathbf{X}_j\|_2^2)$ promotes a more similar original sample of new representations for fusion. The non-negative constant ϕ controls the pressure ratio of the fused terms. when $\phi = 0$ corresponds to unit weights. Since Eq. (8) is a convex function, it is guaranteed to have a globally optimal solution.

When $\lambda = 0$, $\mathbf{W}$ is a unit matrix where the row and column representations are independent of each other. As λ increases, the associated ranks (i.e., corresponding to the same subspace) are simultaneously fused two by two. That is, MDR adaptively recognizes rows and columns corresponding to the same subspace while automatically making the coefficients of rows and columns corresponding to different subspaces approximate to 0. When λ reaches a certain value, the ideal block-diagonal structure emerges automatically without the need to impose any structural a priori and the number of classified brain regions given in advance on $\mathbf{W}$.

B. Self-expression Coefficient Matrix Solving

We optimize the proposed MDR using Generalized Alternating Direction Method of Multipliers (GADMM) [23, 24]. Firstly, the preparatory knowledge and notation definition then the optimization algorithm is presented.

1) Preparatory knowledge and notation definition:

The fMRI data is reduced to an undirected graph $G = (V, E)$ [25], where V consists of n brain regions and E contains edges formed by pairs of fMRI data points based on BOLD signal similarity. Based on the order of occurrence of the indicated pairs in E , the order $l(i, j)$ of the node pair (i, j) , we give the following node association matrix $\mathcal{J}, \tilde{\mathcal{J}}$ as follows:

$$\mathcal{J}_k^{l(i,j)} = \begin{cases} 1, & \text{if } k = i, \\ 0, & \text{otherwise;} \end{cases} \quad \tilde{\mathcal{J}}_k^{l(i,j)} = \begin{cases} 1, & \text{if } k = j, \\ 0, & \text{otherwise,} \end{cases}$$

where $\mathcal{J}_k^{l(i,j)}, \tilde{\mathcal{J}}_k^{l(i,j)}$ represents the k th element in column $l(i, j)$ of $\mathcal{J} \in R^{n \times |E|}$ and column $l(i, j)$ of $\tilde{\mathcal{J}} \in R^{n \times |E|}$, respectively, where $|E|$ is the number of edges in the graph G . Given a coefficient matrix $\mathbf{W} \in R^{n \times n}$ and the graph G , we define $\mathcal{Q} = \mathcal{J} - \tilde{\mathcal{J}}$. The column difference matrix of $\mathbf{W}$ is $\mathcal{B}_{col}(\mathbf{W}) = \mathbf{W}\mathcal{Q}$, and the row difference matrix is $\mathcal{B}_{row}(\mathbf{W}) = \mathcal{Q}^T \mathbf{W}$, where

$\mathcal{B}_{col}: R^{n \times n} \rightarrow R^{n \times |E|}$ and $\mathcal{B}_{row}: R^{n \times n} \rightarrow R^{n \times |E|}$ are linear operators. As a result, Equation (8) can be rewritten as:

$$\min_{\mathbf{W}} \frac{1}{2} \|\mathbf{X} - \mathbf{X}\mathbf{W}\|_F^2 + \lambda([\mathcal{B}_{col}(\mathbf{W})]_{col,2}\eta + \eta^T[\mathcal{B}_{row}(\mathbf{W})]_{row,2}), \quad (9)$$

where η is the edge weight column vector, and the point-to-point order (i, j) is kept the same as in E. In addition to this, for the matrix $\mathbf{M} \in R^{p \times q}$, we define $[\mathbf{M}]_{col,2} = [\|\mathbf{M}_{\cdot 1}\|_2, \dots, \|\mathbf{M}_{\cdot q}\|_2]$ and $[\mathbf{M}]_{row,2} = [\|\mathbf{M}_{1 \cdot}\|_2, \dots, \|\mathbf{M}_{p \cdot}\|_2]^T$.

2) Optimization Algorithm:

The proposed MDR is next optimized using GADMM with the following procedure:

By introducing auxiliary variables $\mathbf{U}_1$ and $\mathbf{U}_2$, Eq. (9) is first transformed into:

$$\begin{aligned} \min_{\mathbf{W}} \frac{1}{2} \|\mathbf{X} - \mathbf{X}\mathbf{W}\|_F^2 + \lambda([\mathbf{U}_1]_{col,2}\eta + \eta^T[\mathbf{U}_2]_{row,2}), \\ \text{s.t. } \mathcal{B}_{col}(\mathbf{W}) = \mathbf{U}_1, \\ \mathcal{B}_{row}(\mathbf{W}) = \mathbf{U}_2. \end{aligned} \quad (10)$$

It can be solved using Augmented Lagrange Multipliers (ALM):

$$\begin{aligned} \min_{\mathbf{W}, \mathbf{U}_1, \mathbf{U}_2, \Gamma_1, \Gamma_2} \frac{1}{2} \|\mathbf{X} - \mathbf{X}\mathbf{W}\|_F^2 + \lambda([\mathbf{U}_1]_{col,2}\eta + \eta^T[\mathbf{U}_2]_{row,2}) \\ + \text{tr}(\Gamma_1^T(\mathcal{B}_{col}(\mathbf{W}) - \mathbf{U}_1)) + \text{tr}(\Gamma_2^T(\mathcal{B}_{row}(\mathbf{W}) - \mathbf{U}_2)) \\ + \frac{\mu_1}{2} \|\mathcal{B}_{col}(\mathbf{W}) - \mathbf{U}_1\|_F^2 + \frac{\mu_2}{2} \|\mathcal{B}_{row}(\mathbf{W}) - \mathbf{U}_2\|_F^2, \end{aligned} \quad (11)$$

where $\Gamma_1 \in R^{n \times |E|}$ and $\Gamma_2 \in R^{|E| \times n}$ are Lagrange multipliers and $\mu_1, \mu_2 > 0$ are two penalty parameters. We use Augmented Lagrange Multipliers to solve (11) to achieve efficient optimization.

Step 1: Fix the other variables and update $\mathbf{W}$.

To update $\mathbf{W}$, the following objective function must be optimized

$$\begin{aligned} f(\mathbf{W}) = \frac{1}{2} \|\mathbf{X} - \mathbf{X}\mathbf{W}\|_F^2 + \text{tr}(\Gamma_1^T(\mathcal{B}_{col}(\mathbf{W}) - \mathbf{U}_1)) \\ + \text{tr}(\Gamma_2^T(\mathcal{B}_{row}(\mathbf{W}) - \mathbf{U}_2)) + \frac{\mu_1}{2} \|\mathcal{B}_{col}(\mathbf{W}) - \mathbf{U}_1\|_F^2 \\ + \frac{\mu_2}{2} \|\mathcal{B}_{row}(\mathbf{W}) - \mathbf{U}_2\|_F^2. \end{aligned} \quad (12)$$

Let $\tilde{\mathbf{U}}_1 = \mathbf{U}_1 + \frac{1}{\mu_1} \Gamma_1$, $\tilde{\mathbf{U}}_2 = \mathbf{U}_2 + \frac{1}{\mu_2} \Gamma_2$, (12) can be reduced to:

$$\begin{aligned} f(\mathbf{W}) = \frac{1}{2} \|\mathbf{X} - \mathbf{X}\mathbf{W}\|_F^2 + \frac{\mu_1}{2} \|\mathcal{B}_{col}(\mathbf{W}) - \tilde{\mathbf{U}}_1\|_F^2 \\ + \frac{\mu_2}{2} \|\mathcal{B}_{row}(\mathbf{W}) - \tilde{\mathbf{U}}_2\|_F^2. \end{aligned} \quad (13)$$

To achieve an efficient update of $\mathbf{W}$, following [24], we introduce the quadratic operator $\Xi = \frac{1}{2}(\alpha\|\mathbf{W}\|_F^2 - \mu_1\|\mathcal{B}_{col}(\mathbf{W})\|_F^2 - \mu_2\|\mathcal{B}_{row}(\mathbf{W})\|_F^2)$. Specifically, the update for $\mathbf{W}$ is given by

$$\mathbf{W} = \text{argmin} f(\mathbf{W}) + \Xi(\mathbf{W} - \mathbf{W}^k), \quad (14)$$

where $\mathbf{W}^k$ refers to the value obtained from the previous iteration. It is important that the chosen α ensures that the operator $\Xi(\mathbf{W})$ is positive definite. In fact, as stated in [26], the minimum effective value for α must satisfy $\alpha > 2\rho(\sigma(Q))$, where $\sigma(Q)$ represents the largest singular value of matrix Q .

By setting the gradient of $f(\mathbf{W}) + \Xi(\mathbf{W} - \mathbf{W}^k)$ with respect to $\mathbf{W}$ to zero, we can derive the update for $\mathbf{W}$ in each iteration:

$$\mathbf{W} = (\mathbf{X}^T\mathbf{X} + \alpha\mathbf{I})^{-1}(\alpha\mathbf{W}^k + \mathbf{X}^T\mathbf{X} + \mu_1\mathbf{A} + \mu_2\mathbf{B}), \quad (15)$$

where $\mathbf{I}$ is the unit matrix, $\mathbf{A} = (\mathbf{U}_1 + \frac{1}{\mu_1}\Gamma_1 - \mathbf{W}^k Q^T)Q^T$, $\mathbf{B} = Q(\mathbf{U}_2 + \frac{1}{\mu_2}\Gamma_2 - \mathbf{W}^k Q^T)$.

Step 2: Fix the other variables and update $\mathbf{U}_1$.

To update $\mathbf{U}_1$, it is necessary to optimize the following objective function:

$$\begin{aligned} f(\mathbf{U}_1) = \lambda[\mathbf{U}_1]_{col,2}\eta + \text{tr}(\Gamma_1^T(\mathcal{B}_{col}(\mathbf{W}) - \mathbf{U}_1)) \\ + \frac{\mu_1}{2} \|\mathbf{U}_1 - \mathcal{B}_{col}(\mathbf{W})\|_F^2. \end{aligned} \quad (16)$$

According to [24], the update of $\mathbf{U}_1$ is obtained via proximal mapping:

$$\begin{aligned} \mathbf{U}_1 = \arg \min_{\mathbf{U}_1} \frac{1}{2} \left\{ [\mathbf{U}_1 - \mathcal{B}_{col}(\mathbf{W})]_{col,2,2} + \frac{\lambda}{\mu_1} [\mathbf{U}_1]_{col,2} \right\} \\ = \text{prox}_{\sigma_{col}\|\cdot\|} \left(\mathcal{B}_{col}(\mathbf{W}) - \frac{1}{\mu_1} \Gamma_1 \right), \end{aligned} \quad (17)$$

where $\sigma_{col} = \left[\frac{\lambda}{\|\mathbf{U}_{1 \cdot 1}\|_2}, \dots, \frac{\lambda}{\|\mathbf{U}_{1 \cdot |E|}\|_2} \right]$. For matrix $\mathbf{M} \in R^{p \times q}$, we

define $[\mathbf{M}]_{col,2,2} = [\|\mathbf{M}_{\cdot 1}\|_2^2, \dots, \|\mathbf{M}_{\cdot q}\|_2^2]$. The proximal mapping $\text{prox}_{\sigma_{col}\|\cdot\|}(\mathbf{M})$ is applied to each column of $\mathbf{M}$.

Step 3: Fix other variables to update $\mathbf{U}_2$.

To update $\mathbf{U}_2$, we minimize the following objective function:

$$\begin{aligned} f(\mathbf{U}_2) = \lambda\eta^T[\mathbf{U}_2]_{row,2} + \text{tr}(\Gamma_2^T(\mathcal{B}_{row}(\mathbf{W}) - \mathbf{U}_2)) \\ + \frac{\mu_2}{2} \|\mathbf{U}_2 - \mathcal{B}_{row}(\mathbf{W})\|_F^2. \end{aligned} \quad (18)$$

Like the update of $\mathbf{U}_1$, the update of $\mathbf{U}_2$ is also achieved through proximal mapping.

$$\begin{aligned} \mathbf{U}_2 = \arg \min_{\mathbf{U}_2} \frac{1}{2} \left\{ [\mathbf{U}_2 - \mathcal{B}_{row}(\mathbf{W})]_{row,2,2} + \frac{\lambda}{\mu_2} [\mathbf{U}_2]_{row,2} \right\} \\ = \text{prox}_{\sigma_{row}\|\cdot\|} \left(\mathcal{B}_{row}(\mathbf{W}) - \frac{1}{\mu_2} \Gamma_2 \right), \end{aligned} \quad (19)$$

where $\sigma_{row} = \left[\frac{\lambda}{\|\mathbf{U}_{2 1 \cdot}\|_2}, \dots, \frac{\lambda}{\|\mathbf{U}_{2 |E| \cdot}\|_2} \right]^T$, and for $\mathbf{M} \in R^{p \times q}$,

$[\mathbf{M}]_{row,2,2} = [\|\mathbf{M}_{1 \cdot}\|_2^2, \dots, \|\mathbf{M}_{p \cdot}\|_2^2]^T$. The proximal mapping $\text{prox}_{\sigma_{row}\|\cdot\|}(\mathbf{M})$ is applied to each row of $\mathbf{M}$.

Step 4: Fix the other variables and update Γ_1 .

The Lagrange multiplier matrix Γ_1 is updated using the following equation:

$$\Gamma_1 = \Gamma_1 + \mu_1(\mathbf{U}_1 - \mathcal{B}_{col}(\mathbf{W})). \quad (20)$$

Step 5: Fix other variables and update Γ_2 .

Similarly, Γ_2 is updated as:

$$\Gamma_2 = \Gamma_2 + \mu_2(\mathbf{U}_2 - \mathcal{B}_{row}(\mathbf{W})). \quad (21)$$

IV. EXPERIMENT

A. Data Processing

During this paper, we used the NITRC dataset, which provides us with resting-state functional Magnetic Resonance Imaging (fMRI) imaging data containing 48 patients with Mild Cognitive Impairment (MCI) and 45 patients with Normal Controls (NC) recruited by newspaper and media

advertisements, with the age range of these subjects ranging from 67 to 81 years.

During data preprocessing, we utilized the SPM8 toolkit to process all fMRI data. First, to eliminate the effect of signal jitter, we removed the first ten time series of each subject. Then to ensure consistency and comparability of the data, we aligned the remaining images to MNI space and normalized the voxel size to $3 \times 3 \times 3 \text{ mm}^3$. Next we removed the covariates of noisy signals such as cerebrospinal fluid, grey matter white matter and head movements using the Friston model [26], which reduced the interference of these factors in the data analysis. Subsequently, we performed a band-pass filtering process to retain the signals in the 0.01-0.8Hz range, which facilitates the highlighting of low-frequency fluctuations associated with neural activity. Finally, we aligned the preprocessed fMRI data and mapped them to 116 brain regions in the AAL template [27], resulting in a data matrix $X \in \mathbb{R}^{170 \times 116}$ containing 170 time series and 116 brain regions. This data matrix will serve as the basis for subsequent data analyses, helping us to reveal the differences and associations between autistic and healthy children in terms of neural activity.

B. Experimental Settings

To comprehensively evaluate the value of our constructed brain network model in practical applications. We used the accuracy of autism diagnosis to evaluate the model. Because the constructed brain network structure has symmetry, we only extracted the upper triangular elements of the brain network connection matrix to simplify the analysis and reduce the computational complexity. Nevertheless, we still need to further filter the information in the connection matrix. Here, the experimental process in this paper is divided into the following three main steps:

In the initial model construction stage, we used five different methods- Pearson Correlation (PC) [28], Sparse Representation (SR) [29], Low-rank Representation (LR) [30], Sparse and Low-rank Representation (SR+LR) [31], and the MDR studied in this paper-to construct brain networks for all samples. Each of these methods contains a series of hyperparameters, such as threshold size and regular term parameters, the choice of which all have an impact on the quality of the final constructed brain network. Therefore, to evaluate the performance of the models constructed by each method more comprehensively, we constructed the corresponding brain networks using different combinations of parameters for each method.

In the feature extraction stage, to be able to effectively identify the key brain network connectivity features related to autism, we used a t-test for feature extraction. Subsequently, we trained these features using SVM classifiers to construct classification models for autism diagnosis. Since data extraction and classification models can significantly affect the results, and our main objective was to validate the performance of the brain network models, we chose the relatively simple t-test and SVM methods to avoid confounding effects introduced by the complexity of the data extraction and the classification model itself.

In the validation phase of the model, due to the small sample size, leave-one-out cross-validation was employed to validate the model's performance. Specifically, in each loop, we keep one sample as the test set and use the remaining samples for training the model. In this way, it can be ensured that each sample has been used as a test set, thus enabling a comprehensive assessment of the model's ability to generalize across different samples. To optimize the performance of the model, a grid search strategy was used to select the best hyperparameters. For the thresholding of the PC method, we chose the thresholding method of sparsity [5%,10%, 95%,100%], where a threshold of 90% means that about 10% of the spurious or weaker connections are removed. For the regularization parameter λ of the SR method, we chose a sequence of exponential growth from 2^{-5} to 2^5 for the λ parameter of the MDR method, we chose 10 parameter values equally spaced between 2^{-5} and 2^5 . In addition, for solving low-rank constraints, we utilized an optimal search strategy and set the regularization parameter to 10^{-5} to ensure stability and accuracy of the solution.

C. Experimental Results

1) Effectiveness Analysis

We used the constructed brain network for the diagnosis of autism, and we also chose to use accuracy, sensitivity and specificity as the measurements. The results are shown in TABLE 1. The experiments show that the MDR method achieved the best results.

TABLE 1 EFFECTIVENESS OF DIFFERENT METHODS ON MCI CLASSIFICATION PROBLEM ON NITRC DATASET

Methods	Accuracy	Sensitivity	Specificity
PC	48.39	44.44	52.08
SR	52.69	46.67	58.33
LR	46.24	40.00	52.08
SLR	35.48	28.89	41.67
MDR	60.22	53.33	66.67

2) Sensitivity Analysis

Considering the importance of hyperparameters on the results, in the following figure, we give the classification accuracy results of five different methods under 10 sets of hyperparameters. All the results are calculated by leave-one-out cross-validation method:

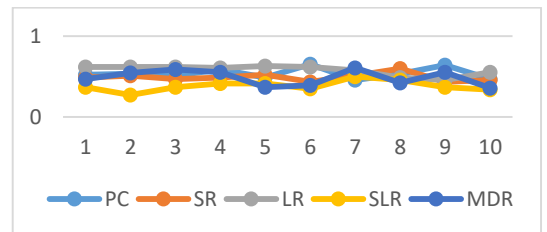


Fig 1 Classification accuracy of 10 different sets of hyperparameters of different methods in the NITRC dataset under leave-one-out cross-validation results.

V. CONCLUSION

In this paper, we propose a new method of modularity-driven regularization (MDR) in biased correlation-based brain network modelling, which significantly improves the accuracy and interpretability of functional connectivity. Unlike previous methods, MDR utilizes a dual $L_{2,1}$ norm to adaptively fuse columns and rows of the coefficient matrix, optimally solved by the GADMM algorithm. This method automatically determines brain network modules without prior assumptions, using a self-expression model to reveal intrinsic connectivity and modular topology with only one hyperparameter, simplifying model tuning. By incorporating module topology, MDR enhances the quality of brain network construction. We applied t-test feature selection and SVM for brain disease classification, validated through nested cross-validation to avoid overfitting and optimize parameters. Compared with existing methods (PC, SR, LR, SLR), MDR significantly enhances brain network classification performance, demonstrating innovation in revealing modular topology and improving model stability and reliability through convex optimization. Overall, the MDR model effectively improves classification accuracy, sensitivity, and specificity in brain disease diagnosis. These results highlight the potential of MDR in improving brain network analysis and its practical application in clinical diagnosis.

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